

EEMB/ES 171
Problem Set #1
Due: October 24, 2025

1. You are studying the effects of climate change on community ecology of ecosystems in central California. You collect the data in Table 1 for a site in the Sierra Foothills. The vegetation is dominated by chaparral and coastal sage. The rains start in the late fall and normally end by the end of April.

Table 1. Climate data.			
Season	Average Temp. (° C)	Rainfall (mm/season)	K for calculating PET
Fall	25	125	1.2
Winter	5	420	1.0
Spring	20	300	1.25
Summer	35	60	1.4

Use the following equation to calculate PET:

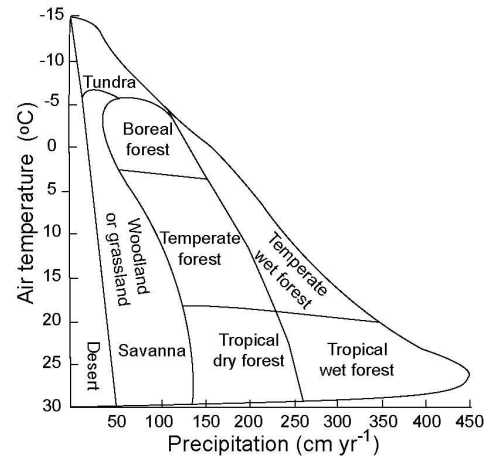
$$\text{PET (mm/season)} = 3 * (15 + (\text{TEMP})^K)$$

K is a constant that depends on the season and is in the table above.

Note: this is not the real formula. I came up with it as a simple equation that predicts plausible numbers. The real equations are complex and depend on latitude.

Assume that the ecosystem can store a total of 100 mm of water in the soil. So, the first rain that falls will go to satisfying PET. After the demands of PET are met, any additional rainfall can go into storage (up to the 100 mm maximum). Water in soil storage will then be available for plant use in the *following* season and can be used to meet the demands of a high PET, and it would then contribute to AET in that season. If the stored water is not needed, it can be kept in storage. Any rainfall in a season that is not used to either a) meet PET demands, or b) banked in the soil, will run off in stream flow.

A. Calculate the water balance for each season. Start with a “water year” that begins with the fall. Assume no water is stored in the soil at the end of the summer. Provide estimates of PET, AET, and stream flow. Also calculate the total annual budget by summing up your seasonal estimates.



B. Describe the moisture conditions for plant growth in each season: Do you expect plants to be actively growing? Merely surviving? Why?

C. Using the biome figure above, what type of biome do you think this would be described as? How sensitive do you think it would be to climate change?

D. You are also studying productivity. So, you model annual net primary productivity (NPP) in this ecosystem. The equation for the global relationship between NPP and AET *for each season* is:

$$\text{NPP (g}\cdot\text{m}^{-2}\text{y}^{-1} \text{ aboveground)} = 1765 * \text{Log (AET+25)} - 3400$$

What would you predict annual NPP to be?

E. Now assume that you actually go out and measure NPP. You get a value of $1200 \text{ g}\cdot\text{m}^{-2}\cdot\text{y}^{-1}$ aboveground. *In case you didn't get the correct answer from part D, this is below the estimated value.* Give **three** reasons why the actual could be lower than the predicted. Your reasons should be based in the biological/ecological controls on productivity, not in something like “the model is wrong”. Then describe, briefly, how you might determine which explanation(s) is correct?

G. Now calculate soil respiration for each season and for the year as a whole.

The model for soil respiration is:

$$\text{Soil respiration} = (M + 5) * e^{(0.075 * T)}$$

M = the **average** amount of moisture stored in the soil over the season (M at beginning + M at the end)/2), not just the moisture at the end.

T = air temperature for the season

*Note that the temperature term is an exponential equation (e to the power $0.075 * T$).*

H. Calculate net ecosystem exchange (NEE) for each season and for the year as a whole.

I. Is this ecosystem gaining or losing carbon?

J. Calculate gross ecosystem productivity (GPP). Assume that plant respiration is equal to NPP.

2. You are studying long term recovery from fire in forest stands in three areas. These areas are all upstream from major urban areas and are used for multiple purposes, including wildlife habitat, tourism, timber production, and supplying the cities with drinking water. You examine recovery by examining stands of different ages post-fire in each area. All three areas have similar vegetation pre-fire, consisting of mature mixed conifers with $1 \text{ Kg} \cdot \text{ha}^{-1} \cdot \text{y}^{-1}$ N export and a net C gain of 5 T/ha. You measure the composition of the plant community, the amount of N being lost in streamflow, and the net accumulation of C in the ecosystem (negative values of C accumulation mean C lost as CO_2 through soil organic matter decomposition). Your results are shown in the table below.

How would list these sites in terms of stability? Consider both resistance and resilience (which is most, which is least)? What criteria did you use to rank these areas? Do this from three perspectives:

- A. The watershed manager, whose concern is supplying her consumers with drinking water.
- B. The land manager who is directly responsible for the use of the lands themselves.
- C. An atmospheric chemist.

Site	TIME SINCE FIRE						
	2	5	10	20	50	100	200
Alpha							
Vegetation	young shrubs	shrubs & hardwood seedlings	Shrubs & hardwood seedlings	juvenile hardwoods	young hardwoods	mature hardwoods and young conifers	mature conifers
N export	40 Kg/ha	25 Kg/ha	5 Kg/ha	1 Kg/ha	1 hg/ha	1 kg/ha	1 kg/ha
C gain	-5 T/ha	10 T/ha	15 T/ha	15 T/ha	10 T/ha	7 T/ha	5 T/ha
Beta							
Vegetation	mostly bare soil with some shrubs	shrubs	Shrubs & conifer seedlings	juvenile conifers	young conifers	mature conifers	mature conifers
N export	40 Kg/ha	25 Kg/ha	15 Kg	5 Kg/ha	1 Kg/ha	1 Kg/ha	1 Kg/ha
C gain	-7 T/ha	6 T/ha	10 T/ha	15 T/ha	12 T/ha	9 T/ha	5 T/ha
Gamma							
Vegetation	grasses	grasses	Grasses & shrubs	shrubs & grasses	Shrubs	shrubs	shrubs
N export	1 Kg/ha	1 Kg/ha	1 Kg/ha	1 Kg/ha	1 Kg/ha	1 Kg/ha	1 Kg/ha
C gain	3 T/ha	10 T/ha	10 T/ha	7 T/ha	5 T/ha	3 T/ha	2 T/ha

